

OPLOGS-07: Analytics & Dashboards

Series: OPLOGS — OpenPipeline Logs | **Notebook:** 7 of 8 | **Created:** December 2025 | **Last Updated:** 07/20/2026

Aggregation, Time Series, and Visualization Queries

This notebook covers aggregation functions, time series analysis, statistical patterns, and dashboard-ready queries for log analytics.

Table of Contents

1. [Aggregation Functions](#)
 2. [Time Series Analysis](#)
 3. [Statistical Analysis](#)
 4. [Dashboard-Ready Queries](#)
 5. [Trend Analysis](#)
 6. [Log Pattern Analysis](#)
 7. [Operational Dashboard Queries](#)
 8.  [Summary](#)
 9.  [Next Steps](#)
 10.  [References](#)
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Prerequisites

-  Access to a Dynatrace environment with log data
-  Completed OPLOGS-01 through OPLOGS-06

-  Familiarity with dashboard concepts

1. Aggregation Functions

DQL Aggregation Functions Reference

Use with summarize command: `summarize {result = function()}, by:{field}`

Counting

- `count()`
Total records
- `countIf(cond)`
Count where condition
- `countDistinct(f)`
Unique values
- `countDistinctIf()`
Unique where cond

Returns: long

Statistical

- `sum(field)`
Total of values
- `avg(field)`
Average value
- `min(field)`
Minimum value
- `max(field)`
Maximum value

Returns: double

Collection

- `collectArray(f)`
All values as array
- `collectDistinct(f)`
Unique values array
- `takeFirst(f)`
First value in group
- `takeLast(f)`
Last value in group

Returns: array

Percentile

- `percentile(f, 50)`
Median (p50)
- `percentile(f, 95)`
95th percentile
- `percentile(f, 99)`
99th percentile
- `stddev(field)`
Standard deviation

Returns: double

Common Aggregation Functions

Function	Description	Example
<code>count()</code>	Count records	<code>summarize {total = count()}</code>
<code>countIf(condition)</code>	Conditional count	<code>countIf(loglevel == "ERROR")</code>
<code>countDistinct(field)</code>	Unique values	<code>countDistinct(k8s.pod.name)</code>
<code>sum(field)</code>	Sum values	<code>sum(bytes)</code>
<code>avg(field)</code>	Average	<code>avg(response_time)</code>
<code>min(field)</code>	Minimum	<code>min(timestamp)</code>
<code>max(field)</code>	Maximum	<code>max(timestamp)</code>
<code>takeFirst(field)</code>	First value	<code>takeFirst(content)</code>
<code>takeLast(field)</code>	Last value	<code>takeLast(content)</code>
<code>collectDistinct(field)</code>	Array of unique values	<code>collectDistinct(loglevel)</code>

Important: Named Aliases Required

```
// ✅ CORRECT – Named aliases
| summarize {count = count(), errors = countIf(...)}, by: {...}
| sort count desc

// ❌ WRONG – Anonymous aggregation can't be used in sort
| summarize count(), by: {...}
| sort count() desc // ERROR!
```

```
// Basic aggregations
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    unique_hosts = countDistinct(dt.entity.host),
    unique_namespaces = countDistinct(k8s.namespace.name),
    unique_pods = countDistinct(k8s.pod.name)
}
```

```
// Aggregations with grouping
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    errors = countIf(loglevel == "ERROR"),
    warnings = countIf(loglevel == "WARN"),
    info = countIf(loglevel == "INFO")
}, by: {k8s.namespace.name}
| sort total desc
| limit 15
```

```
// Error rate calculation
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    errors = countIf(loglevel == "ERROR")
}, by: {k8s.namespace.name}
| filter total > 100 // Minimum sample size
| fieldsAdd error_rate_pct = round((errors * 100.0) / total, decimals: 2)
| sort error_rate_pct desc
| limit 15
```

2. Time Series Analysis

The `makeTimeseries` command creates time-bucketed data for trend visualization.

```
// Log volume over time (5-minute buckets)
fetch logs, from: now() - 6h
| makeTimeseries {log_count = count()}, interval: 5m
```

```
// Error trend over time
fetch logs, from: now() - 6h
| makeTimeseries {
  total = count(),
  errors = countIf(loglevel == "ERROR"),
  warnings = countIf(loglevel == "WARN")
}, interval: 5m
```

```
// Time series by dimension (namespace)
fetch logs, from: now() - 6h
| filter isNotNull(k8s.namespace.name)
| makeTimeseries {
  log_count = count()
}, by: {k8s.namespace.name}, interval: 10m
```

```
// Error rate time series by host
fetch logs, from: now() - 6h
| filter isNotNull(dt.entity.host)
| makeTimeseries {
  total = count(),
  errors = countIf(loglevel == "ERROR")
}, by: {dt.entity.host}, interval: 15m
```

3. Statistical Analysis

```
// Log volume statistics by source
fetch logs, from: now() - 24h
| summarize {
    total = count(),
    first_seen = min(timestamp),
    last_seen = max(timestamp)
}, by: {dt.openpipeline.source}
| fieldsAdd duration_ns = toLong(last_seen) - toLong(first_seen)
| fieldsAdd duration_hours = round(duration_ns / 1h, decimals: 1)
| fieldsAdd logs_per_hour = round(total / (duration_ns / 1h), decimals: 0)
| sort total desc
```

```
// Percentile analysis (if numeric field available)
fetch logs, from: now() - 1h
| fieldsAdd content_length = stringLength(content)
| summarize {
    avg_length = avg(content_length),
    min_length = min(content_length),
    max_length = max(content_length),
    total_logs = count()
}, by: {k8s.namespace.name}
| sort avg_length desc
| limit 15
```

```
// Hourly distribution analysis
fetch logs, from: now() - 24h
| fieldsAdd hour_bucket = bin(timestamp, 1h)
| summarize {log_count = count()}, by: {hour_bucket}
| sort hour_bucket asc
```

```
// Daily distribution analysis
fetch logs, from: now() - 7d
| fieldsAdd day_bucket = bin(timestamp, 1d)
| summarize {
    log_count = count(),
    error_count = countIf(loglevel == "ERROR")
}, by: {day_bucket}
| sort day_bucket asc
```

4. Dashboard-Ready Queries

 Reference queries moved to **OPLOGS-99**. See **OPLOGS-99 § 9.1**

Dashboard-Ready Queries for single-value, time-series, breakdown, and percentile queries optimized for dashboard tiles. The teaching content for *how* dashboard queries are constructed lives in OPLOGS-05 (querying-parsing).

 **Cost-aware design — prefer a metric for queries that run on a schedule.**

Dashboard tiles and alerts re-run their query on every render and evaluation. A `fetch logs ... | summarize` behind a recurring tile pays the Grail **log Query** cost every time, whereas the same answer sourced from a metric does not — the Metrics powered by Grail **Query** dimension is always included at no additional charge, so metric reads never appear on the bill. For any recurring aggregate, check for an out-of-the-box metric first, and otherwise extract one from the log stream at ingest (OpenPipeline metric extraction — see OPLOGS-03 §3). Reserve `fetch logs` for one-shot investigation. **FAQ-09** covers the full decision and the DPS query economics.

5. Trend Analysis

```
// Compare current hour to previous hour
fetch logs, from: now() - 2h
| fieldsAdd period = if(timestamp > now() - 1h, "current", else:
"previous")
| summarize {count = count()}, by: {period, loglevel}
| sort period asc, count desc
```

```
// New error patterns (appeared in last hour)
fetch logs, from: now() - 1h
| filter loglevel == "ERROR"
| fieldsAdd error_sig = substring(content, from: 0, to: 80)
| summarize {
    count = count(),
    first_seen = min(timestamp)
}, by: {error_sig, k8s.namespace.name}
| filter first_seen > now() - 30m // First seen in last 30 minutes
| sort first_seen desc
| limit 15
```

```
// Volume anomaly detection (compare to baseline)
fetch logs, from: now() - 6h
| fieldsAdd time_bucket = bin(timestamp, 15m)
| summarize {log_count = count()}, by: {time_bucket, k8s.namespace.name}
| sort time_bucket desc
| limit 100
```

6. Log Pattern Analysis

```
// Top log patterns (by content prefix)
fetch logs, from: now() - 1h
| fieldsAdd pattern = substring(content, from: 0, to: 60)
| summarize {count = count()}, by: {pattern}
| sort count desc
| limit 25
```

```
// Unique log levels and statuses
fetch logs, from: now() - 1h
| summarize {
    loglevels = collectDistinct(loglevel),
    statuses = collectDistinct(status),
    sources = collectDistinct(dt.openpipeline.source)
}
```

```
// Log diversity score (unique patterns per namespace)
fetch logs, from: now() - 1h
| fieldsAdd pattern = substring(content, from: 0, to: 50)
| summarize {
    total_logs = count(),
    unique_patterns = countDistinct(pattern)
}, by: {k8s.namespace.name}
| fieldsAdd diversity_ratio = round((unique_patterns * 100.0) / total_logs,
decimals: 2)
| sort diversity_ratio desc
| limit 15
```

7. Operational Dashboard Queries

 Reference queries moved to **OPLOGS-99**. See [OPLOGS-99 § 9.2 Operational Dashboard Queries](#) for SLO-style queries: error budgets, top-N services by volume, per-host log distribution.



Summary

In this notebook, you learned:

- ✓ **Aggregation functions** - count, countIf, countDistinct, sum, avg
- ✓ **Time series** - makeTimeseries with intervals
- ✓ **Statistical analysis** - hourly/daily patterns, percentiles
- ✓ **Dashboard queries** - Single values, charts, tables
- ✓ **Trend analysis** - Period comparison, anomaly detection
- ✓ **Pattern analysis** - Log clustering and diversity



Next Steps

Continue to **OPLOGS-08: Security & Data Protection** for data protection patterns.



References

- [DQL Aggregation Functions](#)
 - [DQL makeTimeseries](#)
 - [Dynatrace Dashboards](#)
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